

Theoretical Assignment: Advanced Correlation and Multiple Regression

Instructions: Answer questions independently to demonstrate your conceptual understanding of correlation and regression. While formulas are referenced, the focus is on interpretation and statistical reasoning, not calculation.

1. Covariance vs. Correlation:

The formula for the Pearson product-moment correlation coefficient is often written as $r = \frac{Cov(X,Y)}{S_x S_y}$.

Explain conceptually what the covariance represents and why dividing by the product of the standard deviations ($S_x S_y$) is necessary to interpret the strength of the relationship. Why can't we simply use covariance to compare relationships across different datasets?

2. Least Squares Criterion:

In Ordinary Least Squares (OLS) regression, the regression line is chosen to minimize the Sum of Squared Errors (SSE), represented as $\sum (Y - \hat{Y})^2$.

Why do we square the residuals ($e = Y - \hat{Y}$) rather than simply minimizing the sum of the absolute residuals or the raw residuals? What implication does this have for the influence of outliers on the regression line?

3. Partial Regression Coefficients vs. Zero-Order Correlation:

In a multiple regression model $Y = b_0 + b_1 X_1 + b_2 X_2 + e$, it is possible for the correlation between X_1 and Y (zero-order correlation) to be positive, while the partial regression coefficient b_1 is negative.

Explain how this reversal (suppression effect) can happen. What does b_1 represent that the simple correlation does not?

4. Homoscedasticity and Inference:

One of the key assumptions of OLS is homoscedasticity ($Var(e|X) = \sigma^2$).

If this assumption is violated (heteroscedasticity), the regression coefficients (b) often remain unbiased. However, the standard errors (SE_b) become biased. Explain why biased standard errors are a critical problem for hypothesis testing in social science research. How might this affect Type I or Type II errors?

5. Multicollinearity and Standard Errors:

The variance of a regression coefficient b_j can be expressed as $Var(b_j) = \frac{\sigma^2}{(1-R_j^2) \sum (X_j - \bar{X}_j)^2}$, where $1/(1 - R_j^2)$ is the Variance Inflation Factor (VIF).

Using this formula conceptually, explain why high multicollinearity (high correlation between predictors) leads to "unstable" estimates. What happens to the confidence intervals of your coefficients as VIF increases?

6. **Adjusted R^2 vs. R^2 :**

The formula for Adjusted R^2 is $1 - (1 - R^2) \frac{n-1}{n-k-1}$.

Explain the function of the penalty term $\frac{n-1}{n-k-1}$. Why does adding *any* predictor to a model always increase (or keep constant) the regular R^2 , but can decrease the Adjusted R^2 ? Why is Adjusted R^2 preferred for model comparison?

7. **Dummy Variables and Reference Groups:**

You are examining the effect of Race/Ethnicity (coded: White, Black, Hispanic, Other) on Mental Health Scores. You include three dummy variables in your model: Black, Hispanic, Other, leaving white as the reference group.

The intercept (b_0) in this model has a very specific interpretation. Defend the statement: "The intercept represents the mean Mental Health Score for the White group, assuming all other continuous covariates are zero." How would the interpretation of the coefficient for Black (b_{Black}) change if the reference group were switched to Hispanic?

8. **Global F-test vs. Individual t-test:**

In a multiple regression output, you are given an F-statistic for the overall model and t-statistics for individual predictors. Explain the difference in what these two statistics test. Is it possible to have a significant F-test but no significant individual t-tests? Explain how this might occur.

9. **Omitted Variable Bias:**

Argue why excluding a relevant predictor variable (one that is correlated with both the dependent variable and an included independent variable) results in biased parameter estimates, whereas including an irrelevant variable only results in efficiency loss. Use the concept of "variance partitioning" or "shared variance" in your explanation.

10. **Influential Data Points:**

Distinguish between a data point that has high *leverage* versus one that has high *influence* (e.g., Cook's Distance). Can a data point have an extreme value on the predictor (X) without significantly altering the regression slope? Under what conditions would an outlier warrant removal from a PhD-level analysis?